

Network 1

Section 4

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Section 4:DNS Server

What is a Server?

- A server is a computer that provides services to other computers (clients) over a network.
- Examples:
 - A web server sends web pages.
 - A file server stores files.
 - A DNS server translates names to IP addresses.

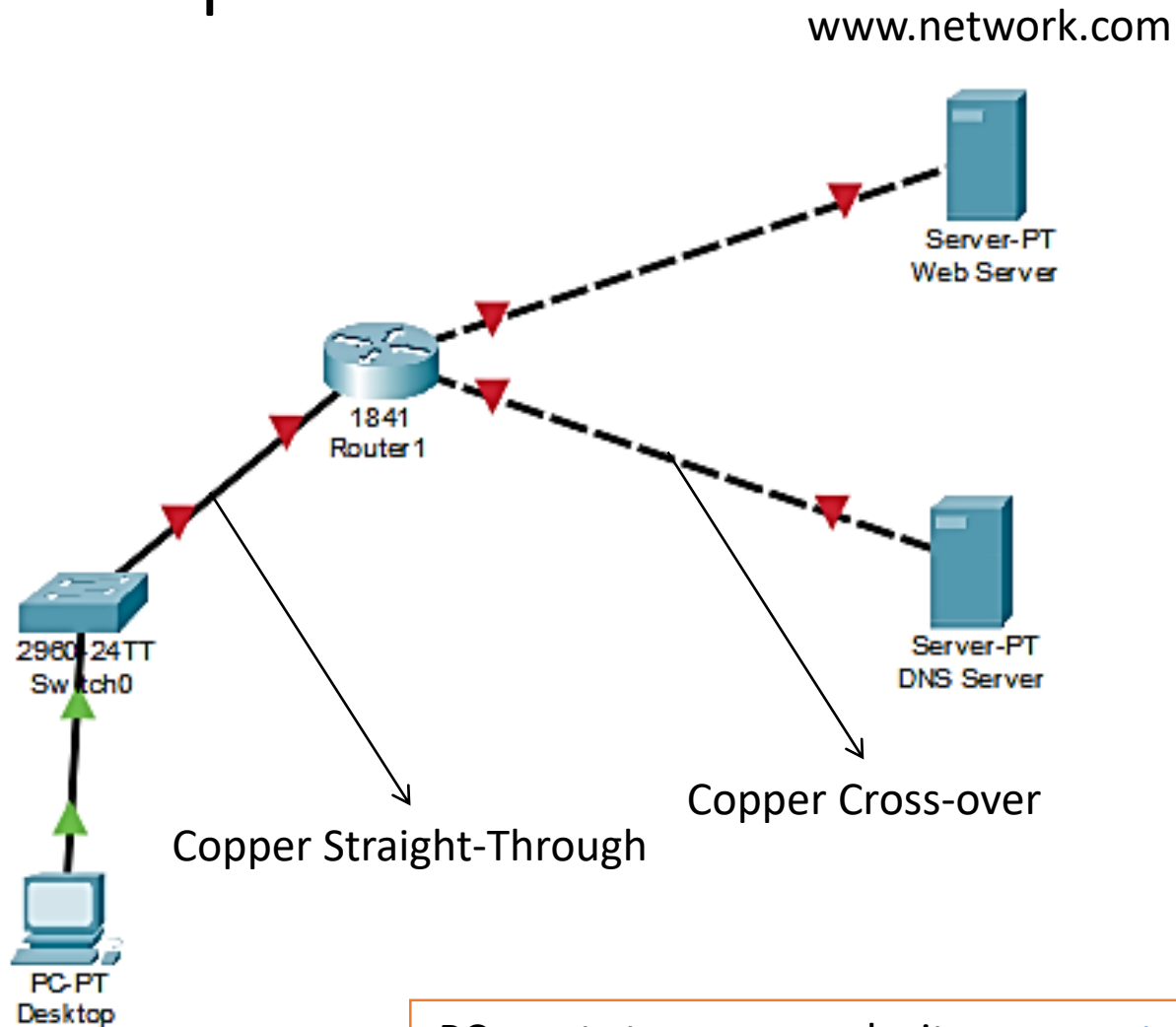


Server

DNS server

- A DNS (Domain Name System) server is a computer/server that stores and provides information about domain names and their corresponding IP addresses so that devices on a network can find each other easily.
- It is a distributed naming system used on the Internet and local networks to map:
 - Domain Name → IP Address
 - www.google.com → 142.250.190.14

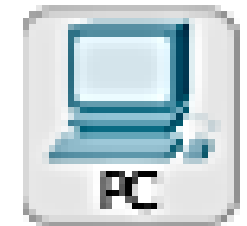
Example



PC wants to access web site www.network.com using domain name



Router



PC

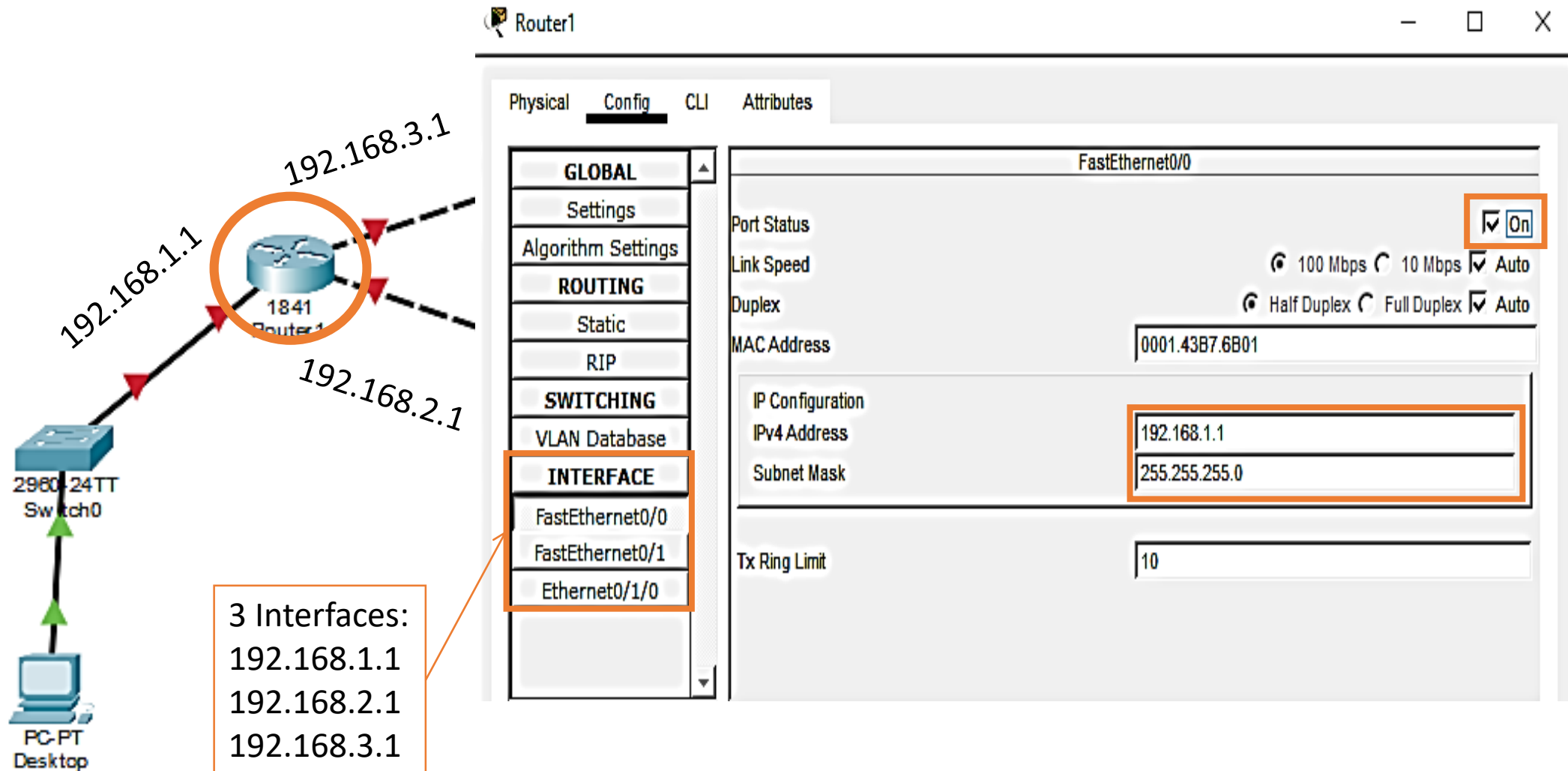


2 Servers

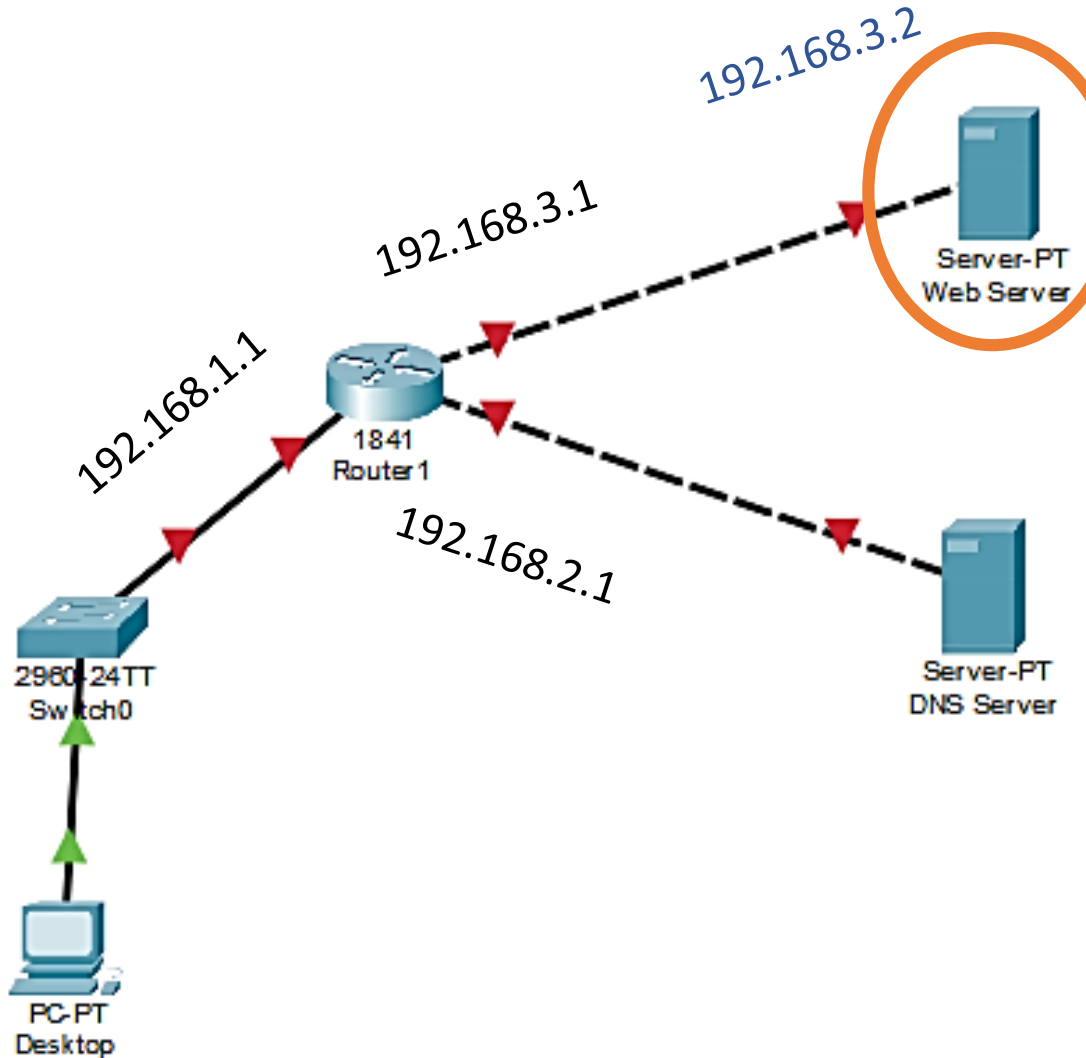


Switch

Router Configuration



Web Server Configuration



Web Server

Physical Config Services **Desktop** Programming Attributes

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.3.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.3.1

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::2E0:F9FF:FE90:8398

Default Gateway:

DNS Server:

802.1X

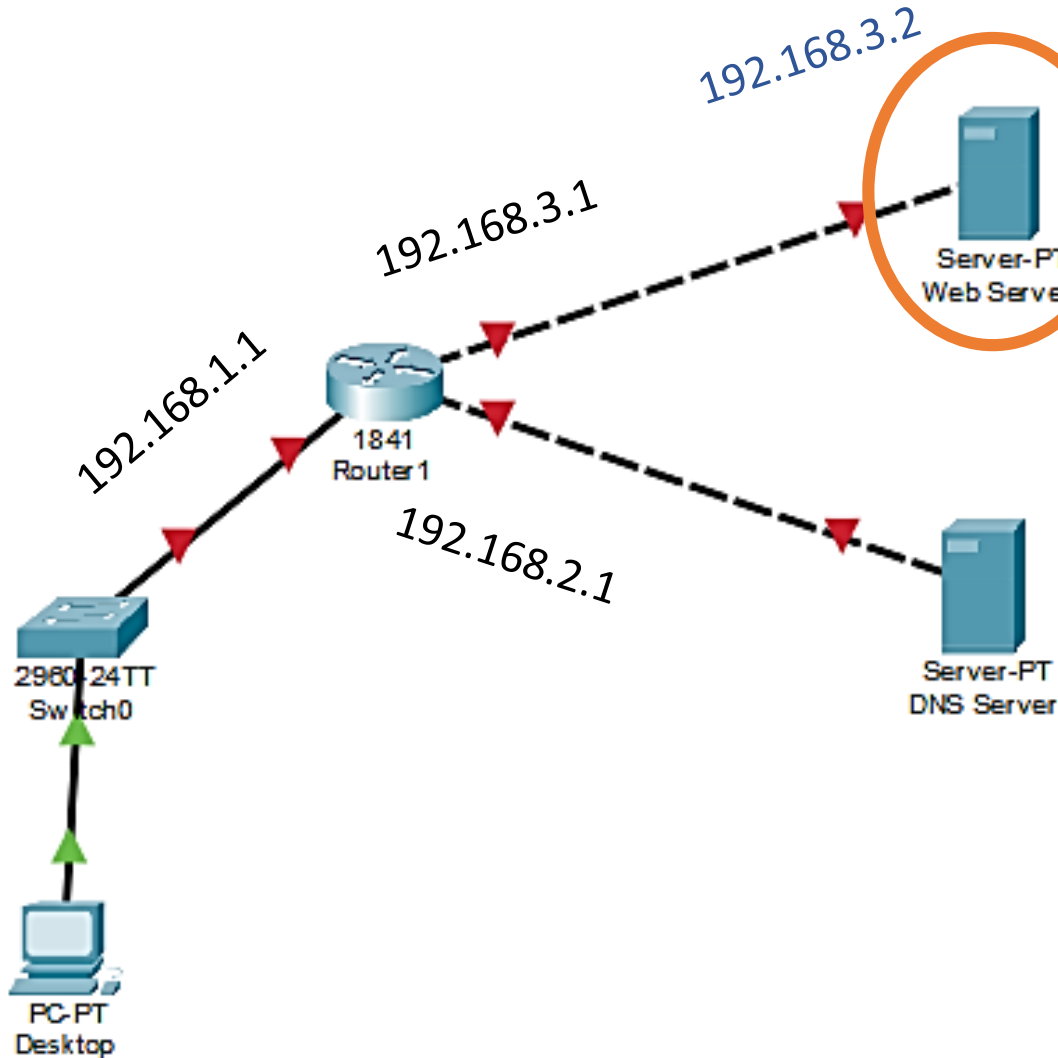
☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

Web Server Configuration



Web Server

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP**
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

HTTP

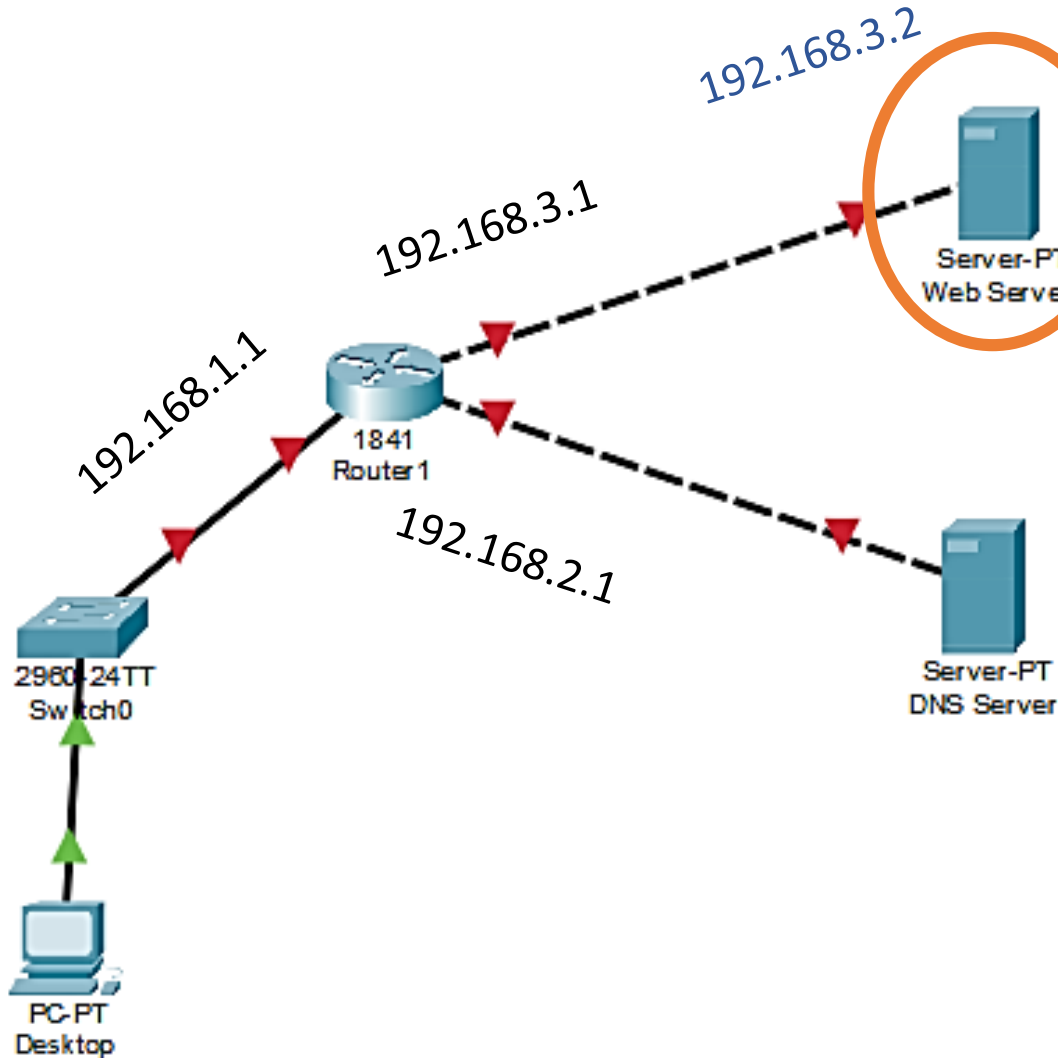
HTTP ☒ On ☐ Off

HTTPS ☒ On ☐ Off

File Manager

	File Name	Edit	Delete
1	copyrights.html	(edit)	(delete)
2	cscoptlogo177x111.jpg		(delete)
3	helloworld.html	(edit)	(delete)
4	image.html	(edit)	(delete)
5	index.html	(edit)	(delete)

Web Server Configuration



Web Server

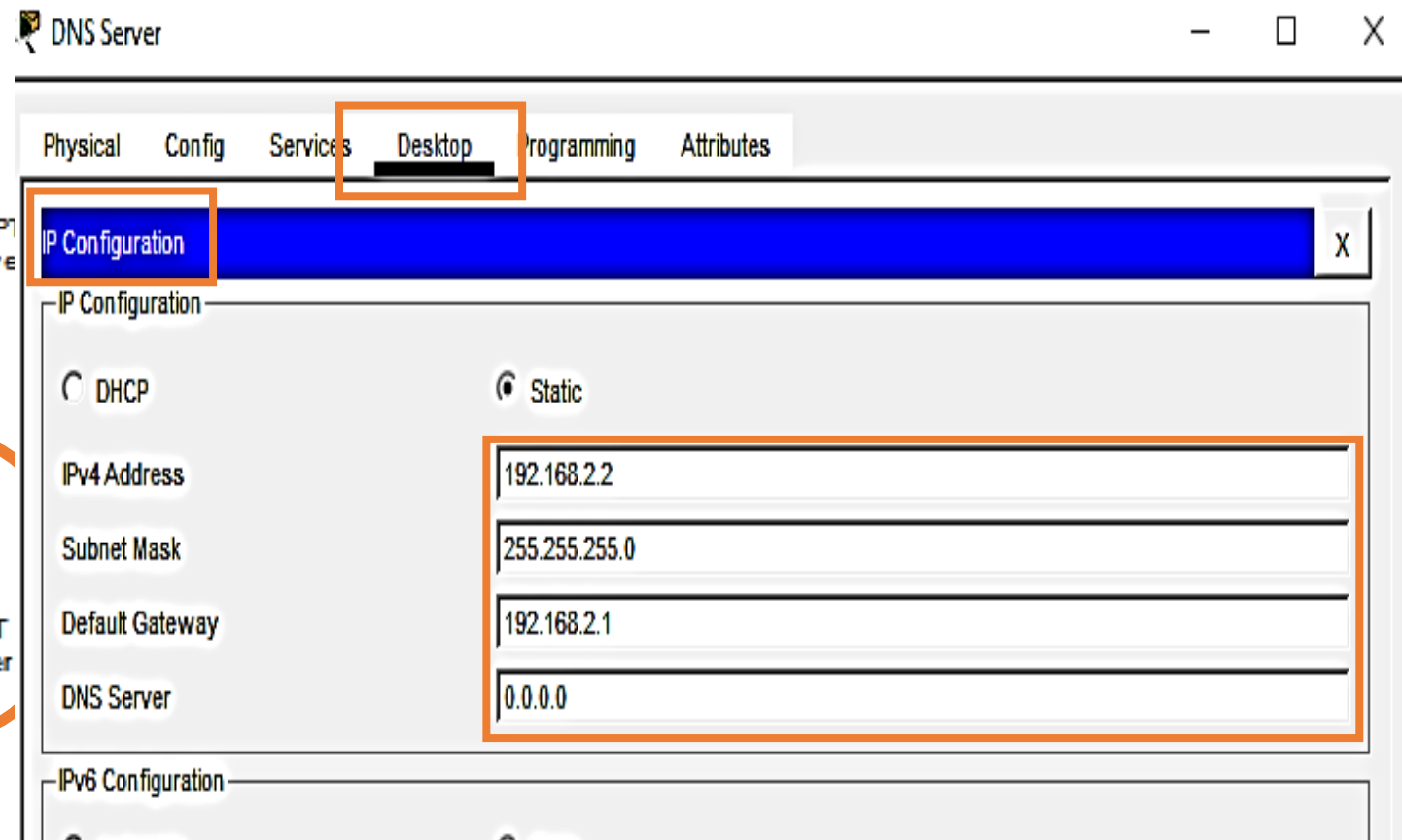
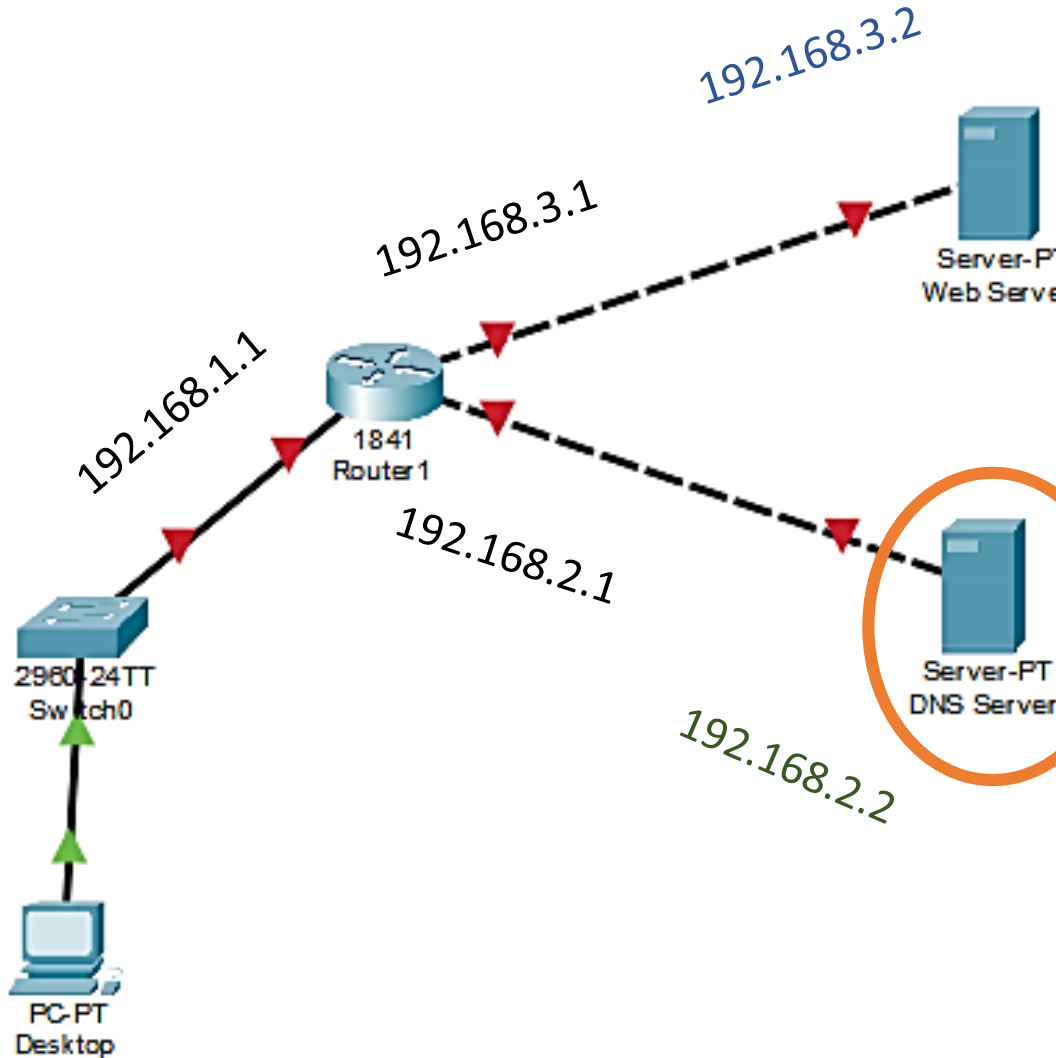
Physical Config **Services** Desktop Programming Attributes

File Name: index.html

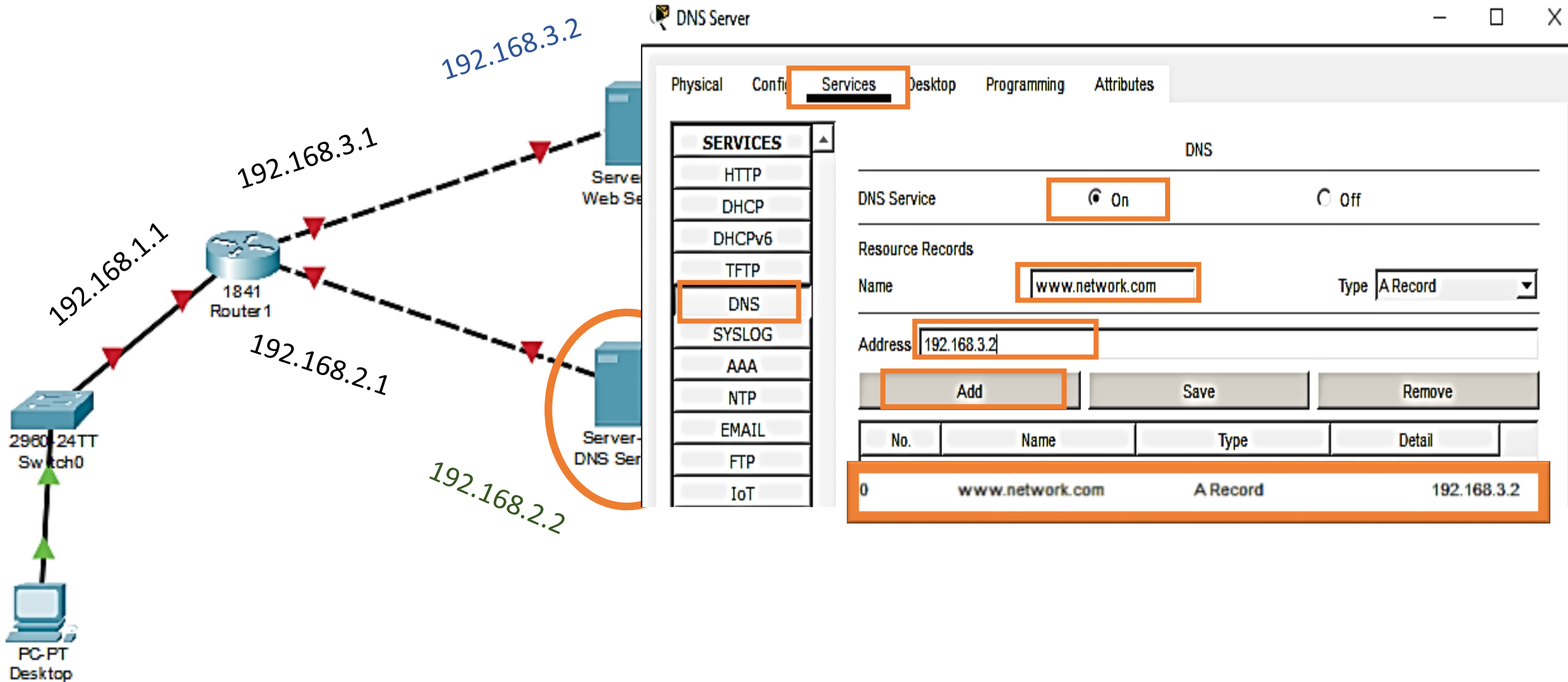
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP

```
<html>
<center><font size='+2' color='blue'>Welcome to
OUR Network</font></center>
</html>
```

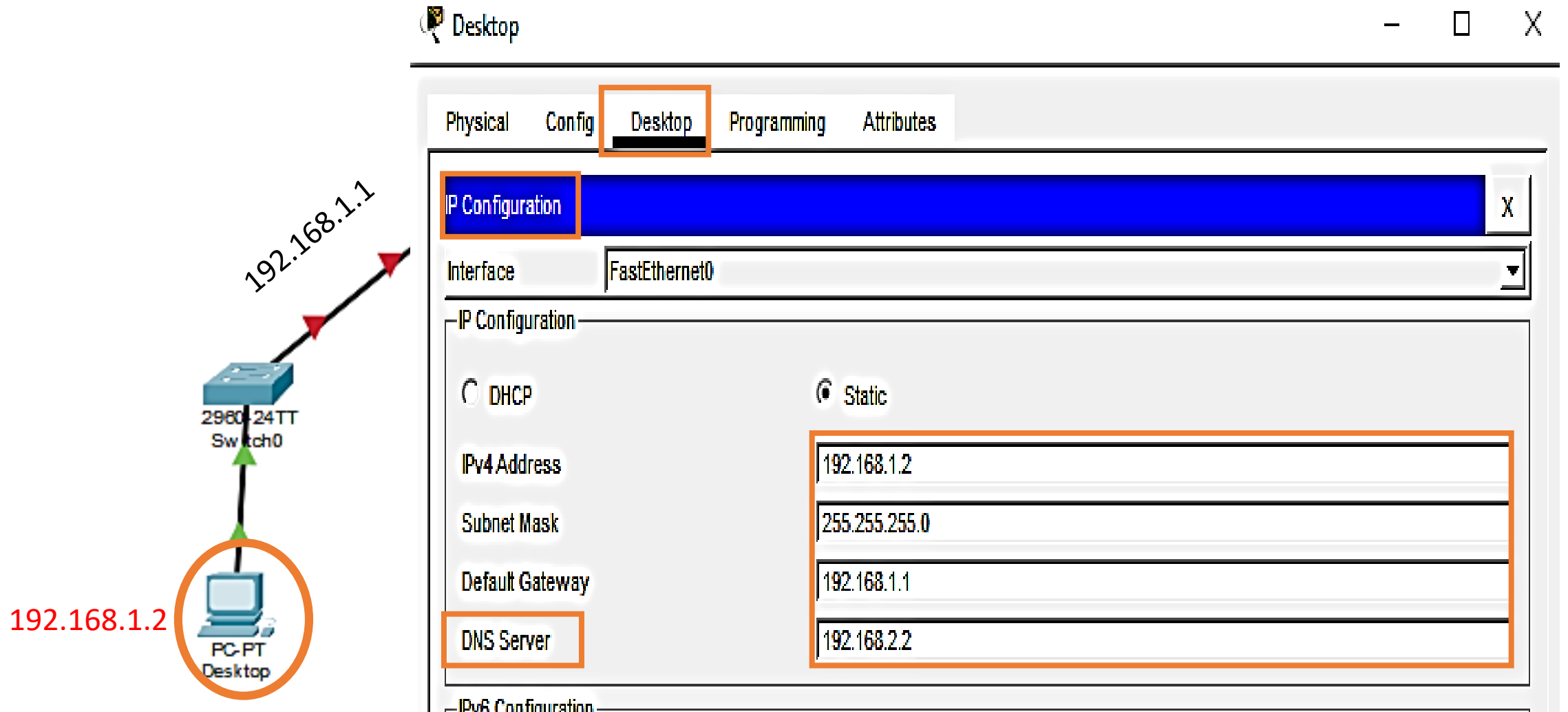
DNS Server Configuration



DNS Server Configuration



PC Configuration



The image shows a network diagram on the left and a PC configuration window on the right. The network diagram features a blue switch labeled "2960-24TT Switch0" connected to a PC icon labeled "PC-PT Desktop". A red arrow points from the switch to the PC, labeled "192.168.1.1". The PC icon is circled in orange, and the IP address "192.168.1.2" is written in red next to it. The configuration window on the right is titled "Desktop" and has tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is selected and highlighted with an orange box. Inside this tab, the "IP Configuration" section is highlighted with a blue bar and an orange box. Below this, the "Interface" is set to "FastEthernet0". The "IP Configuration" section shows two radio buttons: "DHCP" (unselected) and "Static" (selected). Below these are four input fields, each highlighted with an orange box: "IPv4 Address" (192.168.1.2), "Subnet Mask" (255.255.255.0), "Default Gateway" (192.168.1.1), and "DNS Server" (192.168.2.2). The "DNS Server" label is also highlighted with an orange box.

192.168.1.1

2960-24TT Switch0

192.168.1.2

PC-PT Desktop

Desktop

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.2

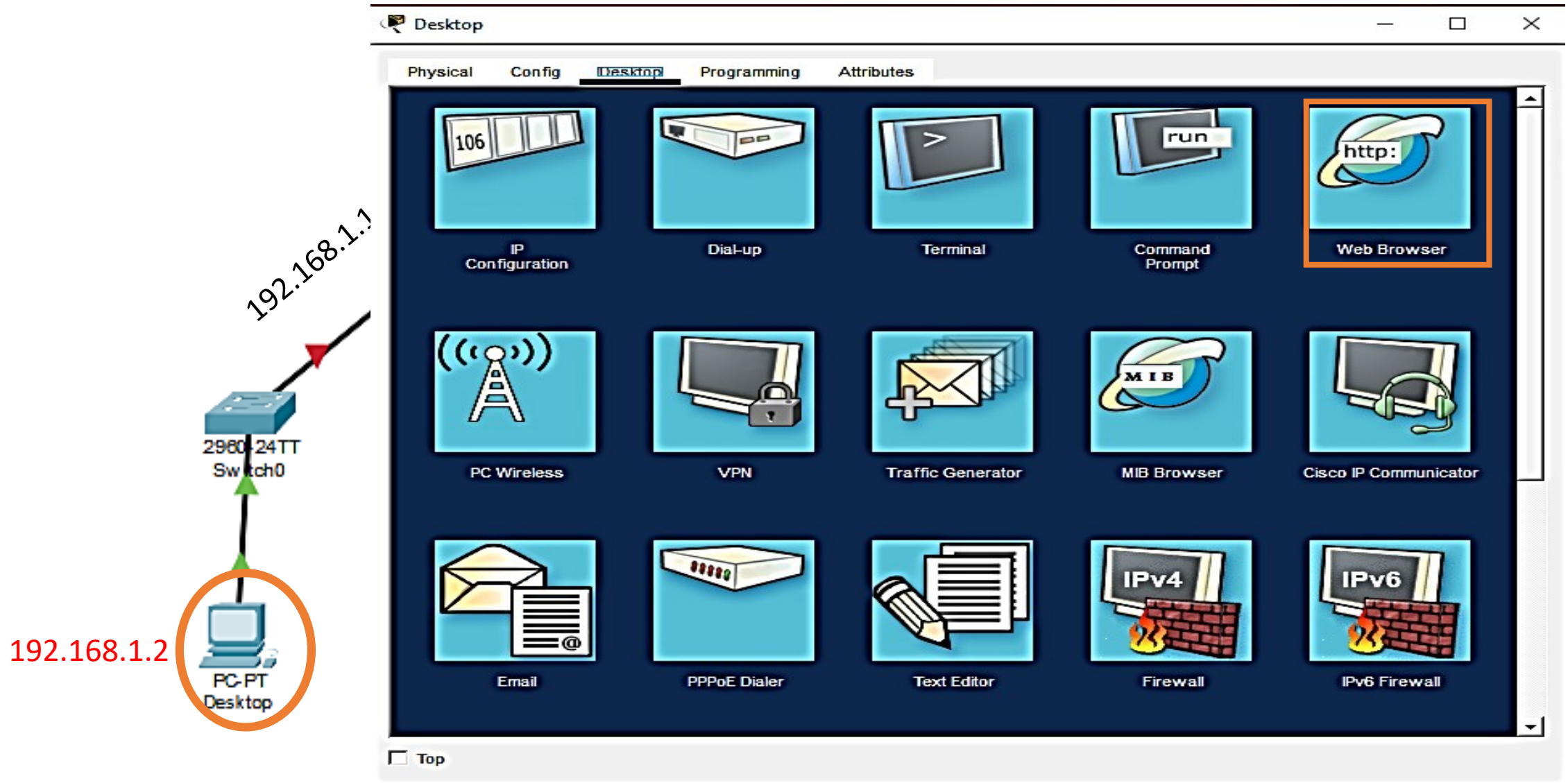
Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 192.168.2.2

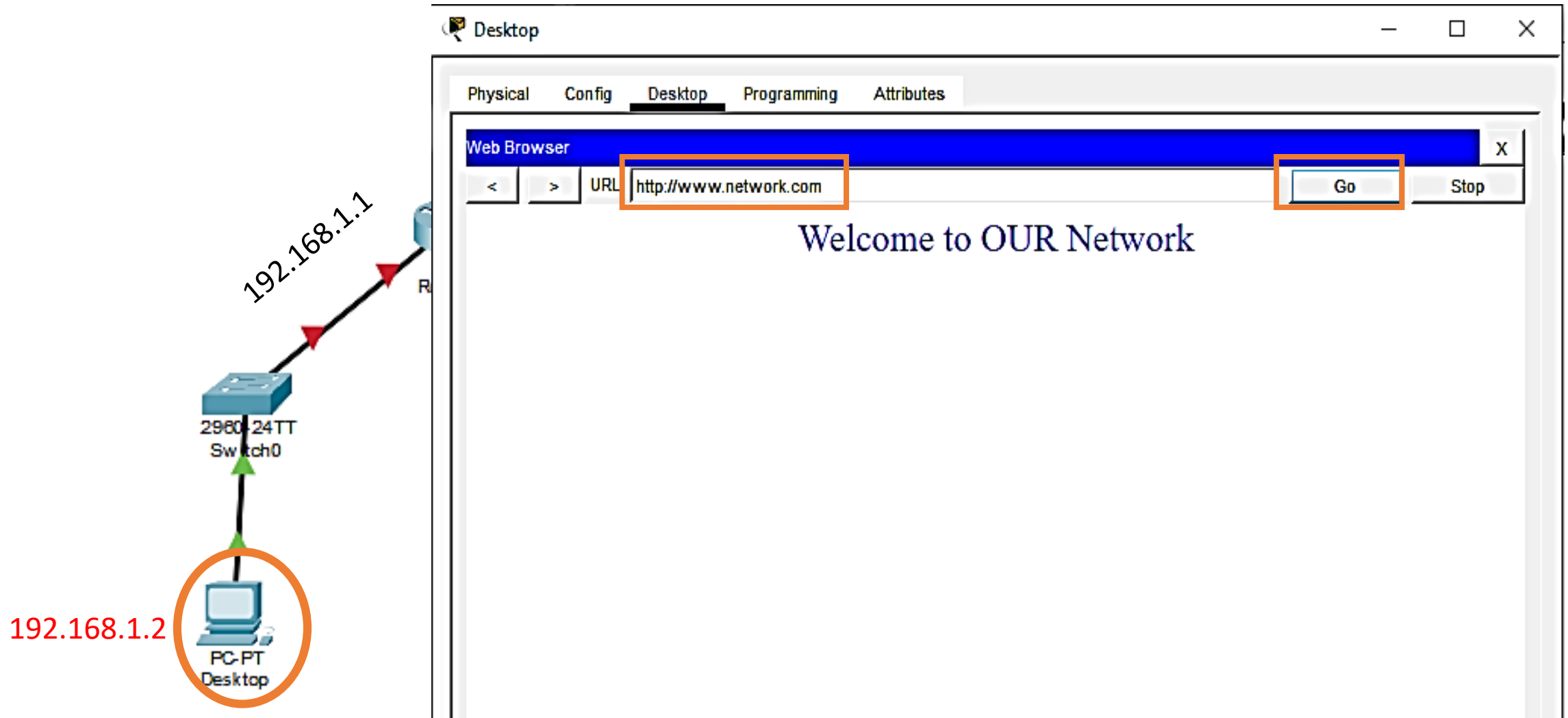
IPv6 Configuration

PC Accessing Website



PC Accessing Website

Web page (index.html)
www.network.com



Any Questions?

Thanks 😊